# Dennis Johan Loevlie

*Multimodal foundation models for tabular data · probabilistic deep learning*

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## EDUCATION

**Centrum Wiskunde & Informatica & University of Amsterdam** — Incoming, Aug 2026 –

PhD in Computer Science, *ELLIS PhD Program*. Advisors: Madelon Hulsebos (TRLab, CWI) and Jan-Willem van de Meent (AMLab, UvA). Focus: multimodal tabular foundation models.

**Tufts University, Medford, MA** — Sep 2024 – May 2026 (expected)

M.S. in Computer Science. GPA: 3.95 / 4.00. Transcript.

**Carnegie Mellon University, Pittsburgh, PA** — Sep 2019 – Dec 2020

M.S. in Chemical Engineering. GPA: 3.91 / 4.00.

**West Virginia University, Morgantown, WV** — Sep 2016 – Aug 2019

B.S. in Chemical Engineering, *cum laude*, with Honors.

## PUBLICATIONS

*Machine Learning*

[1] **2025**. "Synthetic Data Reveals Generalization Gaps in Correlated Multiple Instance Learning". In: *ML4H 2025 Symposium (Findings Track)*.

*Chemistry & Computational Science*

[2] **2023a**. "Demystifying the Chemical Ordering of Multimetallic Nanoparticles". In: *Accounts of Chemical Research* 56.3. [code], pp. 248–257.

[3] **2023b**. "Single Atom Alloys Segregation in the Presence of Ligands". In: *Journal of Physical Chemistry C* 127.46, pp. 22790–22798.

[4] **2022a**. "Size-Dependent Shape Distributions of Platinum Nanoparticles". In: *Nanoscale Advances* 4, pp. 3978–3986.

[5] **2022b**. "Resolving Electrocatalytic Imprecision in Atomically Precise Metal Nanoclusters". In: *Current Opinion in Chemical Engineering* 36, p. 100784.

## RESEARCH EXPERIENCE

**Tufts University — Hughes Lab** — Aug 2024 – Present

*Graduate Researcher with Prof. Michael Hughes*

- Co-led the ML4H 2025 study showing a *generalization gap* between correlated and non-correlated multiple instance learning (MIL) architectures, using a synthetic benchmark with a Bayes-optimal upper bound.
- Identified and resolved a quadratic-scaling bottleneck in the lab's MIL codebase, enabling experiments on substantially larger 3D image datasets.
- Designed attention- and transformer-based MIL pipelines on 3D MRI/CT for early indicators of dementia and stroke; trained on Tufts HPC with multi-GPU distributed setups.
- Developing a regularization method to encourage interpretable attention scores, and methods for supervised learning under noisy labels.

**Tufts University — Sinapov Lab** — Jan 2025 – Present

*Graduate Researcher with Prof. Jivko Sinapov*

- Trained Qwen2.5-Coder-7B with Group Relative Policy Optimization (GRPO) and a custom 3-part reward (structure, aesthetics, semantic alignment) to generate SVGs from text; achieved an **18% improvement** on a benchmark covering aesthetics, alignment, and code validity.
- Investigating why vision-language models (Qwen3-VL) and open-vocabulary segmentation models (SAM 3) fail at object counting in dense scenes; designing tool-use and trajectory-based methods drawing on cognitive-science accounts of human counting.

### University of Pittsburgh — CANELa Lab

Jun 2021 – Jan 2023

*Graduate Researcher with Prof. Giannis Mpourmpakis*

- Proposed a weight-initialization method that delivered a **71% RMSE reduction** for predicting nanoparticle stability across compositions and atomic orderings (*Loevlie et al., Acc. Chem. Res. 2023*).
- Designed and tuned ML architectures for material-property prediction; co-author on *Salem et al., JPCC 2023*.
- Wrote the ML methods and background section for the *Nagarajan et al.* review (Curr. Opin. Chem. Eng., 2022).

### Carnegie Mellon University — Kitchin Group

Dec 2019 – Dec 2020

*Graduate Researcher with Prof. John Kitchin*

- Trained a CNN classifier to extract experimental information from microscopy images; rebuilt Mathematica analysis tooling as fast, interactive Python.
- Released `nb_search`, an open-source Python package for searching and indexing across Jupyter notebooks.

## AWARDS & HONORS

2026 **1st place** (10 teams), *BrainStorm Neural Decoder Challenge* — real-time auditory decoding from 1024-channel ECoG; 95% accuracy, sub-ms inference, edge-deployable model.

2024 **Community Grant**, *Hugging Face* — for Depth Anything demonstration work on video.

2019 **1st place** (Advanced category), *AVEVA National Simulation Competition*.

2018 **2nd place**, *AIChE National Poster Competition*, Computing & Process Control Division.

## INDUSTRY EXPERIENCE

### Massachusetts General Hospital & Harvard Medical School

2025 – Present

*Machine Learning Researcher — computational pathology*

- **Systems:** building segmentation models for colon-cancer pathology slides at gigapixel resolution; pipeline integrates with clinical workflows for downstream evaluation.
- **Research:** dataset-construction methodology and collaboration with pathologists to ground model outputs in expert-validated labels.

### KEF Robotics, Pittsburgh, PA

Jan 2023 – Aug 2024

*Senior Computer Vision & ML Engineer (2024) · Computer Vision Engineer (2023)*

- **Systems:** sole ML engineer on a one-year, $500K project; led a team of five shipping on-device object detection, monocular depth, and 3D map generation; deployed on flight hardware via TensorRT; **45% inference-speed gain** at $<1\%$ accuracy cost. Two in-person demos, selected for follow-on funding.
- **Research:** fine-tuned Mask2Former for power-line segmentation across RGB and IR modalities via transfer learning and cross-modal label transfer.

### ContentsPal

May 2025 – Aug 2025

*Multimodal AI Intern (insurance-tech startup, MIT-led)*

- **Systems:** integrated and tested learning-based duplicate detection and open-vocabulary instance segmentation in React / React Native.
- **Research:** prototyped retrieval and matching components on multimodal product data.

### AiThElite

Dec 2020 – May 2021

*Lead Data Scientist (collegiate-athletics AI startup)*

- Built end-to-end data pipeline (Selenium scrapers → NumPy/Pandas feature engineering → scikit-learn models) and shipped a Django/AWS-hosted prediction product.

## OPEN SOURCE

- `huggingface/transformers` — contributor; merged PRs on model implementations and bug fixes
- `Franblueee/torchmil` — core contributor; multiple merged PRs across the multiple-instance-learning library
- `loevlie/neuropt` — author; LLM-guided neural architecture search and hyperparameter tuning
- `JohanDL/Depth-Anything-Video` — author; Hugging Face Community Grant recipient

## SELECTED TALKS

*Synthetic Data Reveals Generalization Gaps in Correlated Multiple Instance Learning*. Poster, ML4H 2025 Symposium, San Diego, CA. *Dec 2025.*

*Computer Vision for Unmanned Aerial Vehicles*. Invited talk, XChangeIdeas Pittsburgh. *2023.*

## SERVICE & OUTREACH

Industry volunteer, youth robotics team building accessibility tools for blind soccer players 2024

## TECHNICAL SKILLS

**Languages** Python · C++ · JavaScript · MATLAB
**ML / AI** PyTorch · JAX · Hugging Face (Transformers, Datasets, TRL, PEFT) · GRPO / DPO · multimodal learning · RL post-training · multiple instance learning · computer vision
**Engineering** FSDP · DeepSpeed · vLLM · FlashAttention · Triton · CUDA · TensorRT · SLURM / HPC · multi-GPU distributed training · Docker · AWS